

ENVA - Agricultural Encroachment Monitoring Report

Kafr El Dawwar City, Beheira Governorate, Egypt

Report generated: 2026-09-09T19:36:25 UTC
Baseline year: 2016 | Comparison year: 2026
Run ID: ENVA_AGRI_ENCROACH_20260909_181951

1. Executive Summary

Between 2016 and 2026, satellite-derived analysis identified 0.0813 km2 (19.36 feddan) of land within Kafr El Dawwar City that was classified as agricultural in 2016 and as built-up in 2026. This represents 5.68% of the city's 2016 agricultural land base and 1.38% of the total city area. An estimated 4.47% of the city's 2016 agricultural carbon-sequestration potential is associated with this encroached land. 186 discrete encroachment patches were identified and ranked by area for field-verification prioritization.

2. Key Figures

Metric	Value
Precise encroached area	81,347 m2 (0.0813 km2)
Encroached area in feddan	19.36 feddan
% of 2016 agricultural land lost	5.68%
% of total city area	1.38%
Relative carbon-potential loss	4.47% of city 2016 potential
Absolute CO2 loss	Not quantified (no validated local biomass factor)
Discrete encroachment patches identified	186
Monitoring checks logged to date	3
Automated alerts triggered to date	0

3. Methodology

Study area: verified Kafr El Dawwar CITY boundary (not the larger Markaz), reused read-only from the ENVA Smart Environmental Indicators project. Data source: Sentinel-2 Surface Reflectance (COPERNICUS/S2_SR_HARMONIZED) via Google Earth Engine, cloud-masked. Agricultural land is defined as NDVI >= 0.35 and not NDBI-dominant; built-up land uses the same NDBI-dominant rule as the Indicators project's EUDI index, for methodological consistency. Encroachment = agricultural in the 2016 baseline AND built-up in the 2026 comparison year, evaluated pixel-by-pixel at native 10 m resolution (exact computation, not a resampled approximation), with an internal double cross-check confirming the area figure before it is reported anywhere.

4. Top Encroachment Hotspots (Ranked by Area)

Rank	Patch ID	Area (m2)	Feddan	Latitude	Longitude
1	KED-ENCR-0176	14,779	3.52	31.12022	30.13123
2	KED-ENCR-0158	3,416	0.81	31.14078	30.12850
3	KED-ENCR-0085	2,563	0.61	31.12834	30.12206

4	KED-ENCR-0153	2,221	0.53	31.14243	30.12807
5	KED-ENCR-0024	2,050	0.49	31.15015	30.11461
6	KED-ENCR-0143	1,964	0.47	31.14325	30.12758
7	KED-ENCR-0076	1,794	0.43	31.13000	30.12149
8	KED-ENCR-0166	1,538	0.37	31.12058	30.12989
9	KED-ENCR-0102	1,538	0.37	31.13570	30.12282
10	KED-ENCR-0119	1,452	0.35	31.13087	30.12386

5. Continuous Monitoring Status

This project is configured for automated re-checking every 5 days against newly available Sentinel-2 imagery. A check that finds no new valid imagery preserves the last verified result rather than forcing an artificial update. To date, 3 check(s) have been logged and 0 automated alert(s) have been raised on the platform's Early Warning page.

6. Year-by-Year Evidence Timeline (2016-2026)

The table below distinguishes the verified baseline/comparison years and any automated monitoring runs actually present in the evidence ledger. Intermediate years are not fabricated or reconstructed when no verified annual run exists.

Year	Role	Verified evidence	Status / latest verified metric
2016	Baseline	2016 agricultural land + satellite baseline	Verified baseline
2017	Intermediate year	No verified annual run stored	Not analyzed in this run
2018	Intermediate year	No verified annual run stored	Not analyzed in this run
2019	Intermediate year	No verified annual run stored	Not analyzed in this run
2020	Intermediate year	No verified annual run stored	Not analyzed in this run
2021	Intermediate year	No verified annual run stored	Not analyzed in this run
2022	Intermediate year	No verified annual run stored	Not analyzed in this run
2023	Intermediate year	No verified annual run stored	Not analyzed in this run
2024	Intermediate year	No verified annual run stored	Not analyzed in this run
2025	Intermediate year	No verified annual run stored	Not analyzed in this run
2026	Comparison / current	2026 comparison imagery + monitoring outputs	Current cumulative: 0.0813 km2 3 monitoring run(s)

6. Recommendations for Relevant Authorities

1. The detected encroachment represents 5.7% of the city's 2016 agricultural land - a field verification campaign of the top-ranked hotspots (see Section 9) is recommended as a priority action.
2. The single largest detected patch (KED-ENCR-0176, 3.52 feddan) should be prioritized for the first site visit, given it represents the largest continuous land-use change.
3. Encroachment accounts for an estimated 4.5% of the city's 2016 agricultural carbon-sequestration potential - this should be factored into any afforestation/offsetting prioritization (see the ENVA 100 Million Trees initiative).
4. This analysis is a satellite-derived change-detection proxy and does not constitute a legal or administrative determination of illegal encroachment. Field verification and cadastral cross-checking are required before any enforcement or regulatory action.

7. Limitations

- This is a satellite-derived land-cover change proxy, not a legal/cadastral determination of illegal encroachment.
- 2016 Sentinel-2 coverage is sparser than 2026 (satellite launched mid-2015) - baseline confidence is flagged accordingly in the evidence record.
- The carbon-loss tons-CO2 figure uses an uncalibrated generic conversion factor and should not be cited as a validated emissions estimate.
- Field verification is required before any regulatory or enforcement action.

This report was generated automatically by the ENVA Agricultural Encroachment Monitoring system. All figures are traceable to the evidence ledger accompanying this report (consolidated_evidence_ledger.json) and to the underlying satellite imagery and computation logs.